

Cells - Answer Key

1. Prokaryotic cells have a true nucleus and specialized organelles.

Answer: False

Explanation: The textbook states that prokaryotic cells lack a membrane-bound nucleus and most organelles.

2. Glycolysis is an aerobic process.

Answer: False

Explanation: The textbook states that glycolysis is an anaerobic process, meaning it does not require oxygen.

3. Describe the structure and function of the cell membrane, including its role in controlling the movement of substances in and out of the cell.

Answer: The cell membrane, also known as the plasma membrane, is a selectively permeable barrier that separates the cell's interior from its external environment. It is composed primarily of a phospholipid bilayer embedded with proteins, and its structure is described by the fluid mosaic model. The cell membrane controls what enters and exits the cell through passive transport, such as diffusion and osmosis, and active transport, which requires energy in the form of ATP to move substances against their concentration gradient. This regulation is crucial for maintaining the cell's internal environment and allowing it to function properly.

Explanation: This question requires a detailed description of the cell membrane's structure and function, including its role in controlling the movement of substances in and out of the cell.